

MSTAR PRACTICE WORKSHEET**Lesson 6-15: Simplify Algebraic Expressions**

Grade 6 Mathematics · Standard 6.AT.7

These are MSTAR-style questions written for this curriculum to rehearse the format. They are not official Maryland assessment items. MSTAR — Maryland's new state test, first given in spring 2027 — uses the same kinds of questions you see here: selected response, select-all, two-part evidence questions, and written responses.

Part A — Skills Check

One point each.

Question 1 SELECTED RESPONSE — CORRECT: ASimplify $7x + 3x$.

- ☒ A $10x$
- ☐ B $10x^2$
- ☐ C $21x$
- ☐ D $73x$

$7x + 3x = 10x$. Add the coefficients: $7 + 3 = 10$.

Question 2 SELECTED RESPONSE — CORRECT: ASimplify $9n + 4 + 2n$.

- ☒ A $11n + 4$
- ☐ B $11n^2 + 4$
- ☐ C $15n$
- ☐ D $92n + 4$

$9n + 2n = 11n$. The constant 4 has no like term, so it stays. Result: $11n + 4$.

Question 3 SELECTED RESPONSE — CORRECT: A

Which pair are like terms?

- ☒ A $6x$ and $2x$
- ☐ B $6x$ and $6y$
- ☐ C $6x$ and 6
- ☐ D $6x$ and $2x^2$

$6x$ and $2x$ are like terms because they both have the same variable (x) raised to the same power (1).

Question 4 SELECTED RESPONSE — CORRECT: A

Simplify $5a + 3 + 2a$.

- ☒ A $7a + 3$
- ☐ B $7a^2 + 3$
- ☐ C $10a$
- ☐ D $52a + 3$

$5a + 2a = 7a$. The constant 3 stays. Result: $7a + 3$.

Question 5 SELECTED RESPONSE — CORRECT: A

The board shows $3x + 5x$ on one channel. Simplify it.

- ☒ A $8x$
- ☐ B $8x^2$
- ☐ C $15x$
- ☐ D $35x$

$3x$ and $5x$ are like terms, so add the coefficients: $3 + 5 = 8$. The variable x stays, giving $8x$.

Question 6

SELECTED RESPONSE — CORRECT: A

Simplify the track: $4a + 2 + a$.

☒ A $5a + 2$

☐ B $5a^2 + 2$

☐ C $6a$

☐ D $7a$

$4a + a = 5a$ (remember a means $1a$). The 2 has no like term, so it stays. Result: $5a + 2$.

Part B — Test-Format Questions

Scoring notes and distractor rationales below each item.

Question 7**TWO-PART QUESTION****Part A — correct answer: B**

Simplify the expression $8x + 3 + 2x - 1$.

- ☐ A $6x + 2$
- ☒ B $10x + 2$
- ☐ C $10x + 4$
- ☐ D $12x$

The terms $8x$ and $2x$ are like terms, so add their coefficients: $8 + 2 = 10$, giving $10x$. The constants 3 and -1 combine to 2 . The simplified expression is $10x + 2$.

- **A:** The x terms were subtracted. Both are positive, so 8 plus 2 gives $10x$.
- **C:** The constants were added as 3 plus 1 . The second is minus 1 , so they combine to 2 .
- **D:** The constants were folded in with the x terms. 3 and negative 1 are not like terms with $8x$.

Part B — correct answer: A

Which reasoning shows why your answer to Part A is correct?

- ☒ A $8x$ and $2x$ share the same variable part, so add their coefficients to get $10x$. The constants 3 and -1 combine separately to 2 , giving $10x + 2$.
- ☐ B Combine every number in the expression into one coefficient: $8 + 3 + 2 - 1 = 12$, giving $12x$.
- ☐ C Subtract the coefficients of the variable terms: $8 - 2 = 6$, giving $6x + 2$.
- ☐ D Add the coefficients of the variable terms and then add both constants: $10x + 4$.

Only terms with the same variable part can be combined, and constants combine with constants. The constant 1 is being subtracted, so $3 - 1 = 2$, and the x terms are added, not subtracted.

Question 8 SELECT ALL THAT APPLY — CORRECT: A, B, C, E

Select ALL statements that are true about the expression $5y + 7 + 3y + y$.

- ☒ A $5y$, $3y$, and y are like terms.
- ☒ B The coefficient of the term y is 1.
- ☒ C The simplified expression is $9y + 7$.
- ☐ D The simplified expression is $16y$.
- ☒ E 7 is a constant and cannot be combined with the y terms.
- ☐ F The simplified expression is $8y + 7$.

All three variable terms have the same variable part y , so they are like terms. A term written as y has an understood coefficient of 1, so $5 + 3 + 1 = 9$, giving $9y$. The constant 7 has no variable and stays separate, so the simplified expression is $9y + 7$. The answer $8y + 7$ leaves out the coefficient 1, and $16y$ wrongly folds the constant into the variable terms.

Part C — Show Your Thinking

Model answer and scoring guidance.

Question 9 WRITTEN RESPONSE · 2 POINTS

Three students simplified $4x + 3 + 2x + 5 + x$ differently. Student A got $7x + 8$. Student B got $7x^2 + 8$. Student C got $6x + 9$. Who is correct? Explain the mistakes the other two students made.

Model answer: Student A is correct: $4x + 2x + x = 7x$ (the lone x counts as $1x$) and $3 + 5 = 8$, giving $7x + 8$. Student B added the coefficients correctly but changed x into x^2 , which combining like terms never does, and Student C left out the lone x and instead pulled a 1 into the constants, landing on $6x + 9$.

Award 2 points for a complete explanation with correct mathematics, 1 point for a correct answer with thin reasoning, 0 for an unrelated response.

Answer Key. Score written responses with the rubric and guidance above; award partial credit per the 1-point rows.